Linear Regression Assumptions

2026-04-17

Introduction

Linear regression rests on four classical assumptions — linearity, independence, homoscedasticity, and Normality of residuals — each with characteristic diagnostics and remedies when violated. Distinguishing the four is essential because their failures have very different consequences: non-linearity biases coefficients, dependence deflates standard errors, heteroscedasticity gives unbiased coefficients with wrong SEs, and Normality matters mainly for small-sample inference. A responsible analysis checks all four and reports the verdict explicitly.

Prerequisites

A working understanding of linear regression, residuals, and the standard regression diagnostics.

Theory

- **Linearity**: $\mathbb{E}[Y \mid X] = \beta_0 + \sum \beta_j X_j$. Failure produces biased coefficients regardless of sample size.
- **Independence**: residuals uncorrelated with each other (and with predictors). Time-series, repeated-measures, or clustered data violate this; the consequence is downwardly biased standard errors and inflated Type I error.
- **Homoscedasticity**: $\mathrm{Var}(\varepsilon \mid X) = \sigma^2$ constant across predictor space. Failure produces unbiased coefficients but wrong standard errors; the fix is robust (sandwich) SEs or weighted least squares.
- **Normality of residuals**: needed for exact small-sample t-test and confidence-interval validity. The central limit theorem makes this assumption mild for large n, where coefficient sampling distributions are approximately Normal regardless of residual distribution.

Assumptions

These four are themselves the assumptions. Diagnostics aim to detect violations, not to "prove" the assumptions hold.

R Implementation

```r
library(car); library(performance); library(lmtest)

fit <- lm(mpg ~ wt + hp, data = mtcars)

# Residuals-vs-fitted: linearity and homoscedasticity
plot(fit, which = 1)

# Q-Q plot of residuals: normality
plot(fit, which = 2)

# Scale-location: homoscedasticity
plot(fit, which = 3)
```

```r
# Residuals vs leverage: influence
plot(fit, which = 5)

# Formal tests
bptest(fit)     # Breusch-Pagan for heteroscedasticity
shapiro.test(residuals(fit))
dwtest(fit)     # Durbin-Watson for autocorrelation

# performance::check_model combines all
check_model(fit)
```

Output & Results

Four diagnostic plots from `plot.lm()` plus three formal tests: Breusch-Pagan for heteroscedasticity, Shapiro-Wilk for residual Normality, Durbin-Watson for first-order autocorrelation. `performance::check_model()` produces an integrated multi-panel diagnostic with annotations.

Interpretation

A reporting sentence: "Diagnostic plots showed no systematic patterns: residuals-vs-fitted scattered randomly around zero, scale-location was approximately horizontal, and the Q-Q plot showed only mild tail deviation. Formal tests confirmed: Breusch-Pagan $p = 0.41$ (no evidence of heteroscedasticity), Shapiro-Wilk $p = 0.32$ (residuals consistent with Normality), Durbin-Watson 1.95 (no autocorrelation). All four classical assumptions are reasonably supported." Always report what was checked and the verdict.

Practical Tips

- Graphical diagnostics are more informative than formal tests, especially at small samples; formal tests have low power below $n = 50$ and reject too easily above $n = 1{,}000$.
- Heteroscedasticity is common; use robust (HC3) SEs from `sandwich::vcovHC` combined with `lmtest::coeftest` if detected. Alternatively, transform the response or use weighted least squares.
- Clustering or repeated measures invalidate the independence assumption; fit a linear mixed model (`lme4::lmer`) rather than ordinary regression.
- Non-linearity is often fixed by a transformation (log, square root), polynomial terms, or splines. The right fix depends on the substantive functional form.
- Gross departures from Normality at large n typically still allow valid CLT-based inference; the assumption matters most for t-tests and CIs at small n.
- Reporting "all assumptions met" without specifying which were checked and how is uninformative; cite the specific diagnostics used.

R Packages Used

Base R `plot.lm()` and `shapiro.test()`; `lmtest::bptest()` and `dwtest()` for heteroscedasticity and autocorrelation; `car::ncvTest()` and `durbinWatsonTest()` for alternatives; `performance::check_model()` for integrated diagnostics; `sandwich` and `lmtest::coeftest()` for robust SE inference.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI